

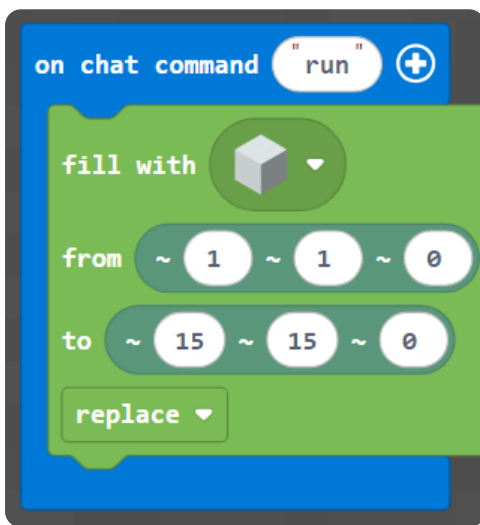
M003 Week 3 — English Worksheet (Beginner)

Topic: Pixel Art on the Wall (x, y) · **Course:** 3D Coordinates · **Level:** Beginner · **Time:** about 30 minutes

This week you **copy a picture made of blocks** onto your wall. Every block sits at a spot named by **two numbers** — (x, y). x is how far **across**, y is how far **up**. You read the picture and place each block at its (x, y).

First, build your canvas. Run this to make a blank 15 × 15 wall, then put a **red block at your feet** as your **home spot**:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Stand on the red block every time you run.** If you move, the picture moves too and the blocks land in the wrong place. Same spot every time = a clean picture.

1 · Predict 🧠

Read the steps. Before you imagine it happening, circle or write your answer.

```
place red block at (6, 1)
place red block at (7, 1)
place red block at (8, 1)
```

Do these blocks make a line going across? Circle one: across · up

Which number stays the same — x or y?

place green block at (8, 13)
place green block at (8, 14)

These share $x = 8$ and y grows. Do they go across or up? Circle one: across · up

2 · Find the Difference 🐛

Each pair shows the **correct** line first, then a **broken** one. Circle what's different and write one short sentence about the bug.

Pair A — A line should grow **across** at the bottom: (5, 1), (6, 1), (7, 1).

```
# correct
place red block at (5, 1)
place red block at (6, 1)
place red block at (7, 1)
```

```
# broken
place red block at (5, 1)
place red block at (5, 1)
place red block at (5, 1)
```

What is wrong? Where do all three blocks land?

Pair B — The stem should grow **up** at $x = 8$.

```
# correct
place brown block at (8, 13)
place brown block at (8, 14)
```

```
# broken
place brown block at (13, 8)
place brown block at (14, 8)
```

What is wrong? Which two numbers got swapped?

3 · Fill the Gap

To move **across** a row, the **first** number (x) has to grow. Fill in the missing numbers so this line goes across the bottom.

```
place yellow block at (6, 1)
place yellow block at (__ , 1)
place yellow block at (__ , 1)
```

Word bank for the blanks: 7 · 8

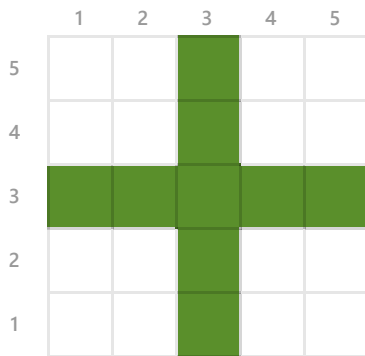
Write the two missing numbers:

4 · Tell Me What You Built 📷

Stand on your **home spot** (the red block).

Warm up first. Copy this small green cross onto your wall. Read each green square's (x, y) and place a green block there.

 Green cross



Now pick one big picture — the 🍍 pineapple or the 🍏 apple from the main worksheet grid — and copy it the same way. Read each colored square and place that color at its (x, y).

When you finish, send a photo of your picture, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

I built the ...

The lowest block in my picture is at (x = ..., y = ...).

One tricky moment was when ...

5 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) while you show your picture on the wall. Talk like you are teaching a friend. Try to use these words: **x, y, across, up, home spot**.

1. Show your home spot (the red block) and say why you stand on it.
2. Show your finished picture.
3. Point at one block and read its coordinate: "this one is at x ..., y ...".

Write what you will say in your video. You can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.